

5.08 Correlation

5.08a	Pearson's product-moment correlation coefficient	a) Be able to calculate the product-moment correlation coefficient (pmcc) for a set of bivariate data; raw data or summarised data may be given. <i>Use of appropriate calculator functions is expected.</i> <i>Learners will not be required to enter large amounts of data into a calculator during the examination.</i>
5.08b		b) Understand that the value of a correlation coefficient is unaffected by linear coding of the variables.
5.08c		c) Understand Pearson's product-moment correlation coefficient as a measure of how close data points lie to a straight line.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
5.08d	Hypothesis tests using Pearson's product-moment correlation coefficient	d) Use and be able to interpret Pearson's product-moment correlation coefficient in hypothesis tests, using either a given critical value, or a p-value and a table of critical values. <i>When using Pearson's coefficient in a hypothesis test, the data may be assumed to come from a bivariate normal distribution.</i> <i>A table of critical values of Pearson's coefficient will be provided.</i>	
5.08e	Spearman's rank correlation coefficient	e) Be able to calculate Spearman's rank correlation coefficient for a maximum of 10 pairs of data values or ranks. <i>Includes being able to draw basic conclusions about the meaning of a value of the coefficient in relation to the ranks before, or without, carrying out a hypothesis test.</i> <i>Includes understanding the conditions under which the use of rank correlation may be appropriate.</i> <i>[Tied ranks are excluded.]</i>	
5.08f	Hypothesis tests using Spearman's coefficient	f) Be able to carry out a hypothesis test for association in a population. <i>Includes understanding that this is a non-parametric test, as it makes no assumptions about the population.</i> <i>Tables of critical values of Spearman's coefficient will be provided.</i>	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
5.08g	Comparison of coefficients	g) Be able to choose between Pearson's product-moment correlation coefficient and Spearman's rank correlation coefficient for a given context. <i>Includes interpreting a scatter diagram and distinguishing between linear correlation and association.</i>	